Agentic Knowledge Graphs for Research Progression

PhD Qualifying Exam Presentation

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Outline

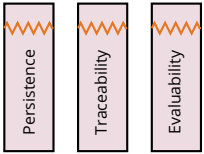
1. Motivation
2. Literature Survey
3. Synthesis
4. Proposal
5. Evaluation
6. Contributions
7. Conclusion

The Structural Problem in AI-Assisted Research

- Research reasoning is **fragmented** and **transient**
- LLM-assisted workflows **amplify volatility**
- **Structural memory** is missing

The Structural Gaps

- **Persistence:** State beyond session boundaries
- **Traceability:** Linkage from Claim $\rightarrow$ Evidence
- **Evaluability:** Empirical measurement of progression

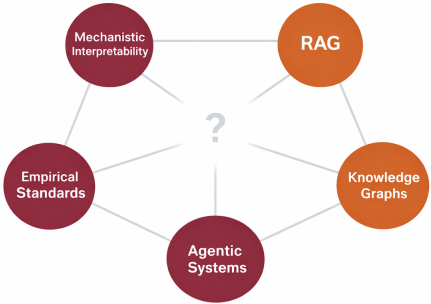


These are infrastructure problems, not model-quality problems.

Guiding Question

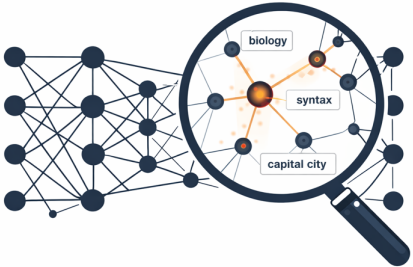
How can research state be made persistent, traceable, and empirically evaluable in AI-assisted workflows?

Landscape Overview



Mechanistic Interpretability

- Reverse-engineer the black box[a]
- Sparse features → human-interpretable structure
- Tells us *how* a model thinks



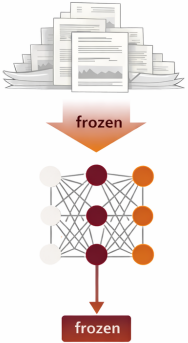
Limitation

[a] Sharkey et al. 2025.

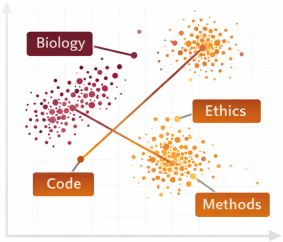
Static weights only — no writable memory for new research state.

Concept-Based Interpretability

Training Data



Latent Space

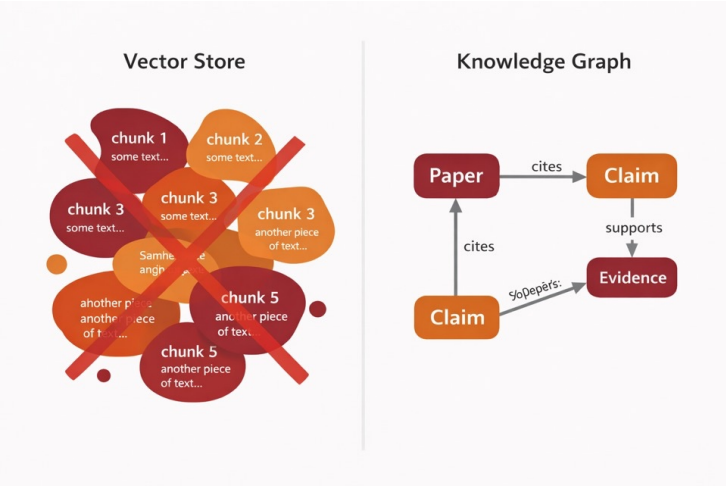


Read-only — no writable memory

Limitation

Read-only — no mechanism to encode new research state.

RAG & Vector Memory Systems



Limitations[a]

[a] Suryawanshi et al. 2025.

- No canonical memory
- No provenance chain
- No structured reasoning
- No longitudinal state

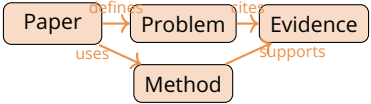
Knowledge Graph Research

Strengths

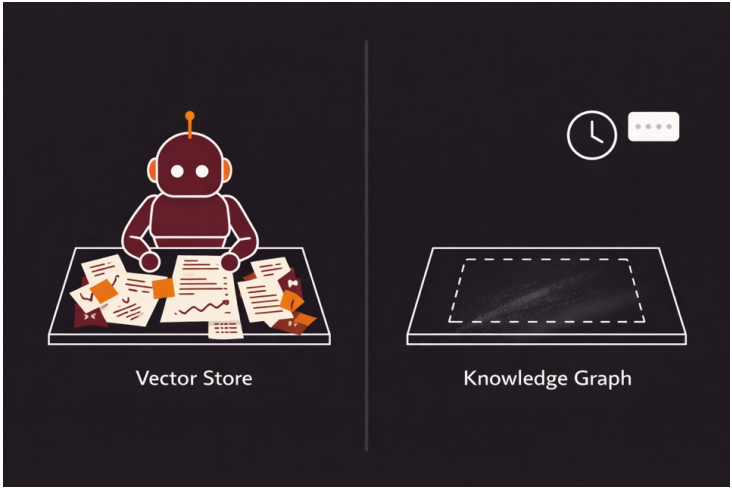
- Ontology-based structure
- Provenance modeling
- Logical relationships

Limitations

- Manual curation
- Static graphs
- No automated maintenance



Agentic Systems



Limitations

- Ephemeral memory
- No structural persistence
- Weak auditability

Empirical Evaluation Standards

- Ralph et al. (2021): Empirical Standards for SE
- Baltes et al. (2025): Guidelines for LLM Studies

Observation

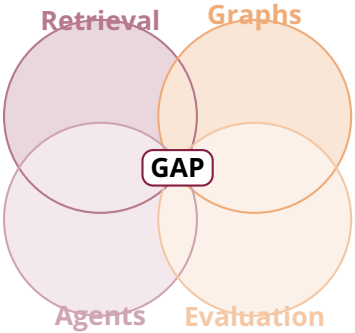
Single-pass or qualitative evaluations dominate.

Need

Longitudinal evaluation of coherent research state over time.

- ☑ Standards defined
- ☑ Guidelines published
- ☐ ~~Agent validation?~~
- ☐ ~~Longitudinal metrics?~~
- ☐ ~~Provenance audit?~~

Cross-Cutting Gaps in the Literature



unstructured

static

amnesic

short-term

This fragmentation motivates new infrastructure.

Positioning My Work

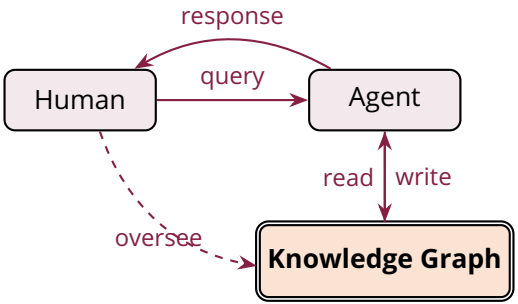
Intersection

Agentic systems + Knowledge graph persistence + Span-level traceability

Literature Refined:

- Need for stronger evaluation standards
- Clearer scope (Software Engineering focus)
- Explicit longitudinal testing

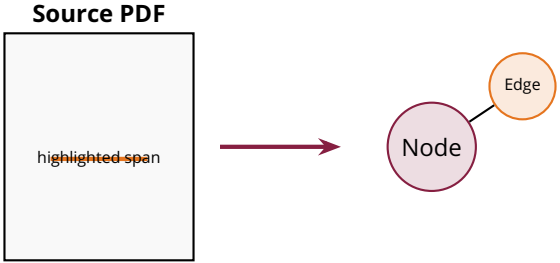
Proposed Architecture



Persistent Research State Loop

- Human-in-the-loop oversight
- Agent-mediated updates

Core Innovation



1. **Span-level evidence alignment** — every node traces to a source sentence
2. **Canonicalization Agent** — de-duplicates entities in the graph
3. **Versioned provenance chains** — full audit trail

Evaluation Framework

Claim	Metric	Baseline
Span-level alignment	Precision / Recall / F1	Standard RAG
Canonicalization	Cohen's κ	Embedding clustering
Traceability	Audit completeness %	RAG w/ citations
Persistence	Retention accuracy	Chat-only LLM

Key Question

Does the system retain knowledge better than a long-context chat window over a 4-week period?

Experimental Design Overview

- **Week 1**
- **Week 2**
- **Week 3**
- **Week 4**

- **Dataset:** Annotated span dataset from SE literature
- **Injection:** Synthetic duplicate injection to test canonicalization

Contributions

1. **Formal definition** of Research State in an AI context
2. **Agentic Curation Protocol** for graph maintenance
3. **Reference infrastructure** implementation

Near-Term Work

- Controlled evaluation studies
- Artifact release (Code & Datasets)
- SE-focused deployment

Long-Term Vision

- Cross-domain expansion (Biology, Physics, CS, . . .)
- Persistent AI-assisted research infrastructure
- Structured memory for scientific discovery

Conclusion

Fragmented Chat → **Persistent State**

- **From:** Fragmented chat interaction
- **To:** Persistent, traceable research state
- **Goal:** Infrastructure for discovery

Questions?

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References I

- Baltes, Sebastian et al. (2025). “Guidelines for Empirical Studies in Software Engineering involving Large Language Models”. In: *arXiv preprint arXiv:2508.15503*. See also llm-guidelines.org.
- Ralph, Paul et al. (2021). “Empirical Standards for Software Engineering Research”. In: *arXiv preprint arXiv:2010.03525*. Version 0.2.0. ACM SIGSOFT Paper and Peer Review Quality Task Force.
- Sharkey, Lee et al. (2025). “Open Problems in Mechanistic Interpretability”. In: *arXiv preprint arXiv:2501.16496*.
- Suryawanshi, Sudhanshu et al. (2025). “A Knowledge Graph-based RAG for Cross-Document Information Extraction”. In: *Proceedings of the 5th International Conference on Pervasive Computing and Social Networking (ICPCSN-2025)*. IEEE. DOI: 10.1109/ICPCSN65854.2025.11035854.